Query

1

```
SELECT DISTINCT directors.director,
   COALESCE(MIN(gross - budget), -1) AS min,
   COALESCE(MAX(gross - budget), -1) AS max,
   COALESCE(TRUNC(AVG(gross - budget), 2), -1) AS avg
FROM directors
   LEFT JOIN movies ON movies.director = directors.director
WHERE date_part('year', CURRENT_DATE) - yearofbirth > 50
GROUP BY directors.director
ORDER BY director,
   min,
   max,
   avg;
```

2

```
SELECT DISTINCT title,
   year
FROM movieawards
WHERE year = (
      SELECT MAX(year)
      FROM (
            SELECT DISTINCT year
            FROM movieawards
            WHERE LOWER(CAST(result AS TEXT)) = 'won'
            GROUP BY (title, year)
            HAVING COUNT(*) >= 3
         ) AS years
   )
   AND LOWER(CAST(result AS TEXT)) = 'won'
GROUP BY (title, year)
HAVING count(*) >= 3
ORDER BY title,
   year;
```

3

```
SELECT DISTINCT title,
   year,
   'most profitable' AS feature
FROM movies
WHERE gross - budget = (
```

```
        SELECT MAX(gross - budget)
        FROM movies
    )
UNION
SELECT title,
    year,
    'least expensive' AS feature
FROM movies
WHERE budget = (
        SELECT MIN(budget)
        FROM movies
    )
ORDER BY title,
    year,
    feature;
```

4

```
SELECT DISTINCT *
FROM(
        SELECT award,
            movieawards.year as year,
            director
        FROM movieawards
            JOIN movies ON movieawards.title = movies.title
            AND movieawards.year = movies.year
        WHERE LOWER(CAST(award AS TEXT)) LIKE '%, best director'
            AND LOWER(CAST(result AS TEXT)) = 'won'
        UNION
        SELECT award,
            year,
            director
        FROM directorawards
        WHERE LOWER(CAST(result AS TEXT)) = 'won'
    ) AS t
WHERE director IN (
        SELECT DISTINCT director
        FROM movies
        WHERE gross > 1000000
            AND date_part('year', CURRENT_DATE) - year <= 5
    )
ORDER BY award,
    year,
    director;
```

5

```
SELECT DISTINCT movies.title,
   movies.year,
   CASE
      WHEN (
         SELECT COUNT(*)
         FROM movieawards
         WHERE movieawards.title = movies.title
            AND movieawards.year = movies.year
      ) = 0 THEN -1
      ELSE TRUNC(
         (
            SELECT COUNT(*)
            FROM movieawards
            WHERE movieawards.title = movies.title
               AND movieawards.year = movies.year
               AND LOWER(CAST(movieawards.result AS TEXT)) = 'won'
         )::decimal /(
            SELECT COUNT(*)
            FROM movieawards
            WHERE movieawards.title = movies.title
               AND movieawards.year = movies.year
         ),
         2
      )
   END success_rate
FROM movies
ORDER BY movies.title,
   movies.year,
   success_rate;
```

6

```
SELECT DISTINCT title,
   year
FROM movieawards
WHERE result = 'won'
GROUP BY (title, year)
HAVING COUNT(*) = (
      SELECT MAX(c)
      FROM(
            SELECT DISTINCT COUNT(*) as c
```

```
            FROM movieawards
            WHERE LOWER(CAST(result AS TEXT)) = 'won'
            GROUP BY (title, year)
        ) AS a
    )
ORDER BY title,
    year;
```

7

```
SELECT DISTINCT *
FROM (
        SELECT director,
            'oldest' AS feature
        FROM directors
        WHERE yearofbirth = (
                SELECT MIN(yearofbirth)
                FROM (
                        SELECT yearofbirth
                        FROM directors
                            JOIN directorawards ON directorawards.director = directors.director
                        WHERE LOWER(CAST(award AS TEXT)) = 'oscar'
                            AND LOWER(CAST(result AS TEXT)) = 'won'
                        UNION
                        SELECT yearofbirth
                        FROM directors
                            JOIN movies ON directors.director = movies.director
                            JOIN movieawards ON movies.title = movieawards.title
                            AND movies.year = movieawards.year
                        WHERE LOWER(CAST(award AS TEXT)) = 'oscar, best director'
                            AND LOWER(CAST(result AS TEXT)) = 'won'
                    ) AS a
            )
        UNION
        SELECT director,
            'youngest' AS feature
        FROM directors
        WHERE yearofbirth = (
                SELECT MAX(yearofbirth)
                FROM (
                        SELECT yearofbirth
                        FROM directors
                            JOIN directorawards ON directorawards.director = directors.director
                        WHERE LOWER(CAST(award AS TEXT)) = 'oscar'
```

```
                    AND LOWER(CAST(result AS TEXT)) = 'won'
                UNION
                SELECT yearofbirth
                FROM directors
                    JOIN movies ON directors.director = movies.director
                    JOIN movieawards ON movies.title = movieawards.title
                    AND movies.year = movieawards.year
                WHERE LOWER(CAST(award AS TEXT)) = 'oscar, best director'
                    AND LOWER(CAST(result AS TEXT)) = 'won'
            ) AS a
        )
    ) AS t
WHERE director IN (
        SELECT director
        FROM directorawards
        WHERE LOWER(CAST(award AS TEXT)) = 'oscar'
            AND LOWER(CAST(result AS TEXT)) = 'won'
        UNION
        SELECT director
        FROM movies
            JOIN movieawards ON movies.title = movieawards.title
            AND movies.year = movieawards.year
        WHERE LOWER(CAST(award AS TEXT)) = 'oscar, best director'
            AND LOWER(CAST(result AS TEXT)) = 'won'
    )
ORDER BY director,
    feature;


8
SELECT CASE
        WHEN (
            SELECT COUNT(*)
            FROM movies
            WHERE year >= 1980
                AND year < 1990
        ) > 0 THEN TRUNC(
            (
                SELECT COUNT(*)
                FROM (
                    SELECT movies.title,
                        movies.year
                    FROM movies
                        JOIN movieawards ON movies.title = movieawards.title
```

```
                    AND movies.year = movieawards.year
                WHERE LOWER(CAST(award AS TEXT)) LIKE 'oscar, %'
                    AND LOWER(CAST(result AS TEXT)) = 'won'
                    AND movies.year >= 1980
                    AND movies.year < 1990
                GROUP BY (movies.title, movies.year)
            ) as a
        )::decimal / (
            SELECT COUNT(*)
            FROM movies
            WHERE year >= 1980
                AND year < 1990
        ),
        2
    )
    ELSE -1
END percentage;
```

9

```
SELECT DISTINCT director
FROM movies
    JOIN movieawards ON movies.title = movieawards.title
    AND movies.year = movieawards.year
WHERE LOWER(CAST(award AS TEXT)) LIKE '%, best director'
    AND gross - budget <= 0
ORDER BY director;
```

10

```
SELECT DISTINCT *
FROM(
        SELECT director
        FROM directorawards
        WHERE LOWER(CAST(result AS TEXT)) = 'won'
        UNION
        SELECT director
        FROM movieawards
            JOIN movies ON movieawards.title = movies.title
            AND movieawards.year = movies.year
        WHERE LOWER(CAST(result AS TEXT)) = 'won'
            AND LOWER(CAST(award AS TEXT)) LIKE '%, best director'
    ) AS c
WHERE NOT EXISTS (
```

```
      SELECT DISTINCT movieawards.year
      FROM movieawards
         JOIN movies ON movieawards.title = movies.title
         AND movieawards.year = movies.year
      WHERE LOWER(CAST(director AS TEXT)) = 'spielberg'
         AND LOWER(CAST(result AS TEXT)) = 'won'
      GROUP BY (movieawards.title, movieawards.year)
      HAVING COUNT(*) >= 3
      EXCEPT
      SELECT year
      FROM (
            SELECT DISTINCT *
            FROM(
                  SELECT director,
                     year
                  FROM directorawards
                  WHERE LOWER(CAST(result AS TEXT)) = 'won'
                  UNION
                  SELECT director,
                     movieawards.year
                  FROM movieawards
                     JOIN movies ON movieawards.title = movies.title
                     AND movieawards.year = movies.year
                  WHERE LOWER(CAST(result AS TEXT)) = 'won'
                     AND LOWER(CAST(award AS TEXT)) LIKE '%, best director'
               ) AS a
            WHERE c.director = a.director
         ) AS b
   )
   AND EXISTS (
      SELECT DISTINCT movieawards.year
      FROM movieawards
         JOIN movies ON movieawards.title = movies.title
         AND movieawards.year = movies.year
      WHERE LOWER(CAST(director AS TEXT)) = 'spielberg'
         AND LOWER(CAST(result AS TEXT)) = 'won'
      GROUP BY (movieawards.title, movieawards.year)
      HAVING COUNT(*) >= 3
   )
ORDER BY director;
```